tilted away from each other by some 60 degrees. No one yet understands why: Some have suggested that we are catching Uranus in a reversal of its north and south magnetic poles, as periodically happens on Earth. Others propose that this too is the consequence of that mighty, ancient collision that knocked the planet over. But we do not know.

Uranus is emitting much more ultraviolet light than it's receiving from the Sun, probably generated by charged particles leaking out of the magnetosphere and striking its upper atmosphere. From a vantage point in the Uranus system, the spacecraft examined a bright star winking on and off as the rings of Uranus passed by. New faint dust bands were found. From the perspective of Earth, the spacecraft passed behind Uranus; so the radio signals it was transmitting back home passed tangentially through the Uranian atmosphere, probing it—to below its methane clouds. A vast and deep ocean, perhaps 8,000 kilometers thick, of super-heated liquid water floating in the air is inferred by some.

Among the principal glories of the Uranus encounter were the pictures. With *Voyager*'s two television cameras, we discovered ten new moons, determined the length of the day in the clouds of Uranus (about 17 hours), and studied about a dozen rings. The most spectacular pictures were those returned from the five larger, previously known moons of Uranus, especially the smallest of them, Kuiper's Miranda. Its surface is a tumult of fault valleys, parallel ridges, sheer cliffs, low mountains, impact craters, and frozen floods of once-molten surface material. This turmoiled landscape is unexpected for a small, cold, icy world so distant from the Sun. Perhaps the surface was melted and reworked in some long-gone epoch when a gravitational resonance between Uranus, Miranda, and Ariel pumped energy from the nearby planet into Miranda's interior. Or perhaps we are seeing the results of the primordial collision that is thought to have knocked Uranus over. Or, just conceivably, maybe Miranda was once utterly destroyed, dismembered, blasted into smithereens by a wild careening world, with many collision fragments still left in Miranda's orbit. The shards and remnants, slowly colliding, gravitationally attracting one